

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 30%

Dr.G.Ayyappan

QUESTIONS

Total Marks: 50

Q.No	Question	Bloom's Level	Marks
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are	BL1 – Remember	5

characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this information about the UPS system.

Course Management System

Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as

- 2
- course with attributes name, department, and c-number
 - section with attributes s-number and enrollment, and dependent as a weak entity set on course
 - room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.

BL2 –
Understand

5

University Registrar's Office Management System

A university registrar's office maintains data about the following entities:

- (a) courses, including number, title, credits, syllabus, and prerequisites;
- (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom;
- (c) students, including student-id, name, and program;
- 3 (d) instructors, including identification number, name, department, and title.
- (e) Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled.

BL2 –
Understand

5

Construct an E-R diagram for the registrar's office.

4	Hospital Management System Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)	BL2 – Understand	5
	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. A department controls a number of PROJECT. Projects have a unique name, number and a single location.		
	Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).		
	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.		
6		BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5

8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand	5
9	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand	5
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

Q1. UPS Product Tracking Information System – Draw an ER Diagram Entities and Attributes

1. Shipped_Item

- Item_No (PK)
- Weight
- Dimensions
- Insurance_Amount
- Destination
- Final_Delivery_Date

2. Retail_Center

- Unique_ID (PK)
- Type
- Address

3. Transportation_Event

- Schedule_Number (PK)
- Type (Flight/Truck)
- Delivery_Route

Relationships

- Retail_Center receives Shipped_Item (1 : M)
- Shipped_Item passes through Transportation_Event (M : N)

Q2. Course Management System – ER Diagram

Entities

Course

- Course_Number (PK)
- Name
- Department

Section (Weak Entity)

- Section_Number (Partial Key)
- Enrollment

Room

- Room_Number (PK)
- Capacity
- Building

Exam

- Time

Relationships

- Course has Section (1:M)
- Section scheduled in Room (M:1)
- Section has Exam (1:1)

Q3. University Registrar's Office Management System – ER Diagram

Entities

Course

- Course_No (PK)
- Title
- Credits
- Syllabus
- Prerequisites

Course_Offering

- Year
- Semester
- Section_No
- Timings
- Classroom

Student

- Student_ID (PK)

- Name
- Program

Instructor

- Instructor_ID (PK)
- Name
- Department
- Title

Relationships

- Course offers Course_Offering (1:M)
- Instructor teaches Course_Offering (1:M)
- Student enrolls in Course_Offering (M:N)
- Enrollment stores Grade

Q4. Hospital Management System – ER Diagram and Tables

Entities

1. Patient

- SS# (PK)
- Name
- Insurance

2. Physician

- Physician_Name (PK)
- Specialization

3. Test_Log

- Test_ID (PK)
- Test_Name
- Date
- Time

4. Patient_History

- History_ID (PK)
- Test_Name
- Date

Relationships

- Physician treats Patient (M:N)
- Patient has Test_Log (1:M)
- Patient has Patient_History (1:M)

Q5. Company Project Management System

Dept_ID (PK)

- Dept_Name
- Location

Employee

- SSN (PK)
- Name
- Address
- Salary
- Sex
- Birth_Date

Project

- Project_No (PK)
- Project_Name
- Location

Dependent

- Dependent_Name
- Birth_Date
- Relationship

Relationships

- Department employs Employees (1:M)
- Department controls Projects (1:M)
- Employee works on Projects (M:N) (Hours per week stored)
- Employee manages Department (1:1) (Manager Start Date stored)
- Employee has Dependents (1:M)
- Employee supervises Employee (1:M)

Q6. Explain Schema and Instance. Distinguish between Logical, Physical and View Schema

Schema

A schema is the overall design or blueprint of a database. It defines the tables, attributes, relationships, and constraints.

Example:

- Student(Student_ID, Name, Department)

The schema remains mostly fixed unless the database structure changes.

Instance

An instance is the actual data stored in the database at a specific point in time.

Example:

Student_ID	Name	Department
101	Rahul	CSE
102	Priya	AI & DS

The instance changes whenever records are inserted, updated, or deleted.

Difference Between Schema Types

Schema Type	Description	Example
Logical Schema	Defines the logical structure of the database, including tables, attributes, and relationships.	Student(ID, Name, Course)
Physical Schema	Describes how data is physically stored on storage devices, including files and indexes.	Data stored in disk blocks with indexing
View Schema	Defines how different users see specific parts of the database.	Faculty can view student marks, while students can view only their own records.

Conclusion:

- Schema = Structure of the database.
- Instance = Actual data stored in the database.
- Logical schema defines the design, physical schema defines storage, and view schema defines the user-specific view of the database.

Q7. Bank Management System – ER Diagram

Entities

Customer

- Customer_ID (Primary Key)
- Name
- Address
- Phone_Number

Account

- Account_Number (Primary Key)
- Account_Type
- Balance

Branch

- Branch_ID (Primary Key)
- Branch_Name
- Location

Loan

- Loan_ID (Primary Key)
- Loan_Amount
- Loan_Type

Relationships

- A Customer owns one or more Accounts (1:M).
- A Branch maintains many Accounts (1:M).
- A Customer can take one or more Loans (1:M).
- A Branch issues many Loans (1:M).

Q8. Explain the Extended ER (EER) Model

The Extended Entity-Relationship (EER) Model is an advanced version of the ER model. It provides additional concepts for designing complex databases.

1. Generalization

Generalization combines two or more lower-level entities into a higher-level entity based on common features.

Example:

- Car and Bike Vehicle

2. Specialization

Specialization divides a higher-level entity into smaller sub-entities based on their

characteristics.

Example:

- Employee Manager, Engineer, Clerk

3. Aggregation

Aggregation treats a relationship as a higher-level entity so it can participate in another relationship.

Example:

An Employee works on a Project. This "Works_On" relationship can be associated with a Department.

Advantages of EER

- Reduces data redundancy.
- Represents complex real-world situations.
- Improves database design.
- Makes relationships easier to understand.

Q9. Hospital Management System – EER Diagram

Superclass

Person

- Person_ID
- Name
- Address
- Phone_Number

Subclasses

Patient

- Patient_ID
- Insurance_Details

Doctor

- Doctor_ID
- Specialization

Weak Entity

Test_Record

- Test_Name
- Date
- Time

(Test_Record depends on Patient for its existence.)

Relationships

- Doctor treats Patient (M:N)
- Patient has Test_Record (1:M)

Perfect. Let's continue with the remaining answers in a detailed, assignment-ready format.

Q7. Bank Management System – ER Diagram

Entities

Customer

- Customer_ID (Primary Key)
- Name
- Address
- Phone_Number

Account

- Account_Number (Primary Key)
- Account_Type
- Balance

Branch

- Branch_ID (Primary Key)
- Branch_Name
- Location

Loan

- Loan_ID (Primary Key)
- Loan_Amount
- Loan_Type

Relationships

- A Customer owns one or more Accounts (1:M).
- A Branch maintains many Accounts (1:M).
- A Customer can take one or more Loans (1:M).
- A Branch issues many Loans (1:M).

Q8. Explain the Extended ER (EER) Model

The Extended Entity-Relationship (EER) Model is an advanced version of the ER model. It provides additional concepts for designing complex databases.

1. Generalization

Generalization combines two or more lower-level entities into a higher-level entity based on common features.

Example:

- Car and Bike Vehicle

2. Specialization

Specialization divides a higher-level entity into smaller sub-entities based on their characteristics.

Example:

- Employee Manager, Engineer, Clerk

3. Aggregation

Aggregation treats a relationship as a higher-level entity so it can participate in another relationship.

Example:

An Employee works on a Project. This "Works_On" relationship can be associated with a Department.

Advantages of EER

- Reduces data redundancy.
- Represents complex real-world situations.
- Improves database design.
- Makes relationships easier to understand.

Q9. Hospital Management System – EER Diagram

Superclass

Person

- Person_ID
- Name
- Address
- Phone_Number

Subclasses

Patient

- Patient_ID
- Insurance_Details

Doctor

- Doctor_ID
- Specialization

Weak Entity

Test_Record

- Test_Name
- Date
- Time

(Test_Record depends on Patient for its existence.)

Relationships

- Doctor treats Patient (M:N)
- Patient has Test_Record (1:M)

Q10. Role of DBA (Database Administrator)

A Database Administrator (DBA) is responsible for managing and maintaining a database system.

Responsibilities

1. Install and configure the DBMS.
2. Create and maintain databases.
3. Manage users and access permissions.
4. Ensure database security.
5. Perform backup and recovery.
6. Monitor database performance.
7. Optimize queries and indexing.
8. Maintain data integrity and consistency.
9. Handle concurrency control.
10. Perform regular maintenance and troubleshooting.

Conclusion

A DBA ensures that the database remains secure, reliable, efficient, and available for all authorized users.

Code of Conduct:

I N.Thejasree (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
N.Thejasree

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application

Integratio
n of
Literature

15%

question/proble
m

Covers all
required parts
of the question

Covers most
parts with
minor
omissions

Covers
only
some
parts

Incomple
te or
missing
respons
e

Clarity

10%

Clear, concise,
and well-
organized
answer

Understanda
b l
e with minor
issues

Somewhat
unclear or
poorly
structured

Difficult to
understand